

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **data architecture and modelling frameworks** for enterprise monitoring dashboards using **Python ETL pipelines, SQL, and Power BI**, consolidating operational metrics across 12+ business systems to enable real-time performance insights and **25% reduction in incident detection time**.
- Defined **AI strategy and mathematical optimization models** for production systems using **scikit-learn, pandas, and statistical analysis** to translate operational inefficiencies into predictive insights, resulting in **18% improvement in system resource utilization** and measurable SLA gains.
- Led **structured data baseline definition and quality assurance** across interconnected enterprise systems via **REST APIs and database-level data flows** using **PostgreSQL, MySQL, Oracle**, ensuring high-fidelity data modelling for downstream AI model training and integration into production platforms.
- Architected **AI-driven operational efficiency solutions** integrated into **Java Spring Boot microservices** running on **AWS** (EC2, Lambda, CloudWatch), automating **intelligent alerting and anomaly detection** workflows that reduced manual triage overhead by **40%** and accelerated mean-time-to-resolution.
- Developed **machine learning models** for time-series forecasting and **supervised/unsupervised learning** using **Python, R statistical methods, and evaluation metrics** to predict system bottlenecks and resource constraints in real-time, enabling proactive optimization across distributed infrastructure.
- Built **automated data pipelines and ETL workflows** using **Python scripting, SQL queries, and CI/CD integration** (Jenkins, Git) to operationalize AI models at scale, ensuring continuous model retraining, versioning, and deployment of optimization logic into production systems with **99.2% uptime**.

SKILLS

AI & Machine Learning – supervised learning, unsupervised learning, predictive modelling, time-series forecasting, anomaly detection, scikit-learn, model evaluation metrics, hypothesis testing.

Data Architecture & Modelling – data foundation design, structured data baselines, data quality assurance, data warehousing, data modelling, ETL pipeline architecture, schema design.

Mathematical Optimization – optimization algorithms, resource allocation, performance optimization, operational efficiency modelling, constraint-based optimization.

Data Engineering & Analytics Stack – Python (pandas, NumPy, scikit-learn), R/RStudio, SQL (PostgreSQL, MySQL, Oracle), Power BI, Tableau, DAX, Power Query, statistical analysis.

Backend & Cloud Infrastructure – Java Spring Boot, REST APIs, AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, CI/CD (Jenkins, Git).

Operational Intelligence & Monitoring – real-time dashboards, anomaly detection, alerting systems, performance metrics, SLA management, incident response, log analysis.

Enterprise Systems & Integration – ERP integrations, cross-system data flows, change-control management, UAT coordination, system interconnectivity.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

Mining Operations Data Intelligence & Optimization Platform (HCL Technologies)

- Designed end-to-end **data foundation and structured baseline** for Bitcoin mining operations, consolidating real-time metrics from distributed mining sites using **Python ETL pipelines, Apache Spark**, and **PostgreSQL data warehousing** to enable unified operational visibility across 50+ geographical locations.
- Developed **predictive ML models** using **scikit-learn** and **time-series forecasting** (ARIMA, Prophet) to optimize hash-rate allocation, power consumption forecasting, and thermal management, achieving **22% improvement in mining efficiency** and **\$1.2M annual cost savings** through intelligent resource distribution.
- Built **real-time anomaly detection and intelligent automation workflows** integrated into core mining platform using **Java Spring Boot microservices** and **AWS Lambda**, automating hardware failure prediction and equipment rebalancing logic that reduced unplanned downtime by **35%** and mean-time-to-detection by **6 hours**.
- Engineered **automated performance dashboards and KPI reporting** using **Power BI** and **DAX measures**, delivering executive-level insights on site-level efficiency, equipment health, and optimization ROI; connected to production systems via **REST API data flows** and **SQL stored procedures** for near-real-time updates.
- Implemented **model training and deployment pipeline** using **Python, CI/CD (Jenkins, Git)**, and **Docker containerization**, enabling continuous optimization model retraining on 48-hour cycles with automated versioning and A/B testing across live mining sites with **zero production disruption**.

Enterprise Payments Operations Monitoring & ISO 20022 Data Platform (HCL Technologies)

- Architected **data modelling framework** for **SWIFT messaging** (MT/MX formats) and **ISO 20022 message flows** using **SQL** and **Python data pipelines**, enabling structured capture of payment transaction metadata across **SEPA, instant payments, and cross-border payment corridors** with full audit trail compliance.
- Designed **operational dashboards** using **Splunk** and **Power BI** for real-time monitoring of payments processing SLAs, transaction-flow visibility, and regulatory alerting, supporting **P1/P2 incident response** with sub-minute detection and **99.8% SLA achievement** for mission-critical banking clients.
- Integrated **machine learning anomaly detection** for fraud and compliance monitoring using **scikit-learn** and **statistical methods**, reducing false-positive alerts by **45%** and enabling proactive identification of suspicious payment patterns aligned with **GDPR, Data Protection, and regulatory compliance** requirements.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)